1. Competency #3: Implement a given strategy to determine a mathematical solution

(Highly relevant)

- Although the formula isn't explicitly stated, the problem tests whether students know and can apply the definition: average speed = total distance ÷ total time.
- A common misconception is to average the two individual speeds; avoiding this error requires correctly implementing the standard strategy.
- This demonstrates strategic knowledge application, not just mechanical computation.

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2. Competency #7: Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

(Highly relevant)

- The solution involves multiple numerical operations:
  - Addition: 4 + 3 = 7 (total distance)
  - Addition: 9 + 6 = 15 (total minutes)
  - Unit conversion: 15 minutes = 0.25 hours
  - Division: 7 ÷ 0.25 = 28
- All steps involve real-world quantitative data, making numerical manipulation central to success.

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3. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process

(Moderately relevant—but critical)

- The context states that the speedometer shows "her average speed for a trip," implying that average speed refers to the entire journey as a single unit.
- Misinterpreting the outbound and return legs as separate "trips" could lead to incorrect reasoning.
- The mention of a "shorter route" also signals that distances differ, reinforcing the need to treat each leg separately when summing total distance and time.
- Thus, contextual awareness helps students correctly model the problem.

4. Competency #1: Perform a simple calculation

(Moderately relevant)

- The arithmetic (addition, division, unit conversion) is straightforward but essential.
- This competency supports higher-level actions but is not the primary focus.

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5. Competency #12: Carry out multi-step procedures

(Moderately relevant)

- The solution follows a clear sequence of steps:
  1. Compute total distance (4 + 3)
  2. Compute total time in minutes (9 + 6)
  3. Convert minutes to hours (15 ÷ 60 = 0.25)
  4. Divide total distance by total time in hours (7 ÷ 0.25)
- While not highly complex, the task requires organized, sequential execution—fitting the description of a multi-step procedure.

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6. Competency #8: Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

(Weakly relevant due to question format)

- As a fill-in-the-blank question, only the final answer (28 km/h) is required; no explanation is needed.
- However, in instructional settings where students must show their work, this competency would become relevant.

1. 能力 #3：实施给定策略以确定数学解

（高度相关）

- 题目未直接给出公式，但考查学生是否掌握“平均速度 = 总路程 ÷ 总时间”这一核心策略。
- 学生需主动调用该策略，而非对各段速度求算术平均（常见错误）。
- 正确解题依赖于对“平均速度”定义的准确理解和应用，体现了策略性知识的实施。

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2. 能力 #7：操作数字、数据、代数表达式等

（高度相关）

- 涉及多项数值操作：
  - 加法：4 + 3 = 7（总距离）
  - 加法：9 + 6 = 15（总分钟）
  - 单位换算：15 分钟 = 0.25 小时
  - 除法：7 ÷ 0.25 = 28
- 所有步骤均围绕真实情境中的数量信息展开，属于典型的数值与单位操作。

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3. 能力 #13：利用对情境的理解来指导或加速数学解题过程

（中等相关，但关键）

- 情境明确指出：“speedometer can tell … her average speed for a trip”，暗示“平均速度”是针对整个行程的指标。
- 若忽略“往返是一个完整行程”的语境，可能误用分段速度平均。
- 此外，“shorter route”提示回程距离不同，强调不能假设对称路径，需分别处理数据。
- 因此，对“trip”和“average speed”在现实工具（速度表）中的含义的理解，直接影响建模正确性。

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4. 能力 #1：执行简单计算

（中等相关）

- 计算本身不复杂（加法、除法、单位换算），但步骤较多，需细心。
- 属于支撑性能力，服务于更高阶的策略实施（#3）和数据操作（#7）。

5. 能力 #12：执行多步骤程序

（中等相关）

- 解题包含清晰的多个步骤：
  1. 求总距离（4 + 3）
  2. 求总时间（9 + 6 分钟）
  3. 将分钟转换为小时（15 ÷ 60）
  4. 用总距离除以总时间（单位为小时）
- 虽非高度复杂，但要求有序、连贯地完成一系列操作，符合“多步骤程序”的特征。

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6. 能力 #8：清晰表述解法，展示中间结果

（弱相关，因题型限制）

- 本题为填空题，仅需填写最终答案（28 km/h），不要求写出过程。
- 但在教学或形成性评价中，若要求学生展示计算步骤，则此项能力会被激活。